

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF ELECTRICAL ENGINEERING
MID TERM EXAMINATION-OCTOBER 2024

Course: B.Tech

Semester: III

Subject Name: NETWORK ANALYSIS

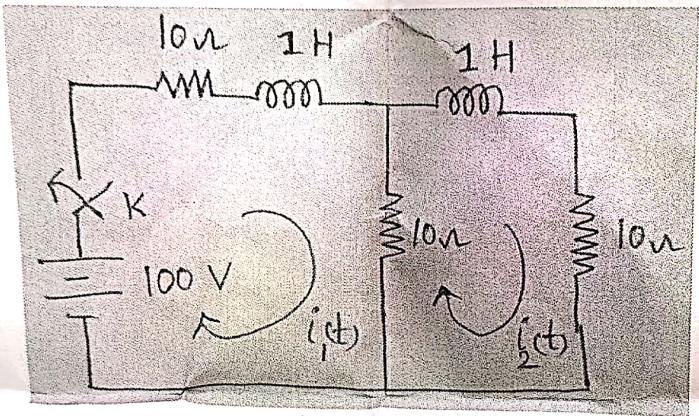
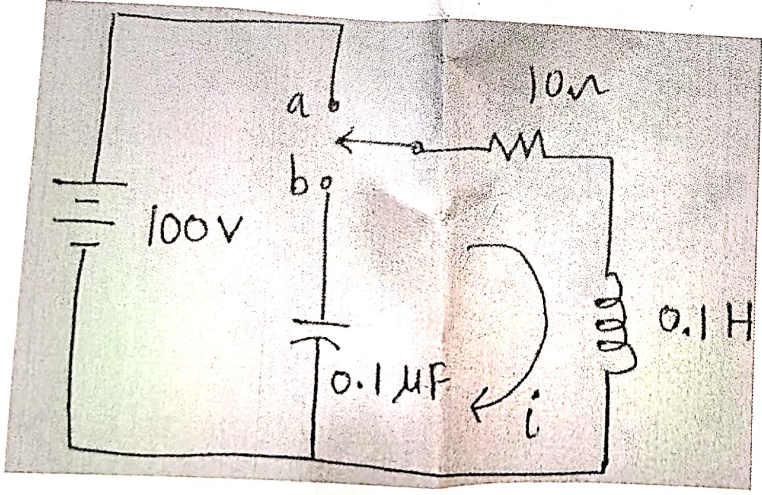
Subject Code: EE-213

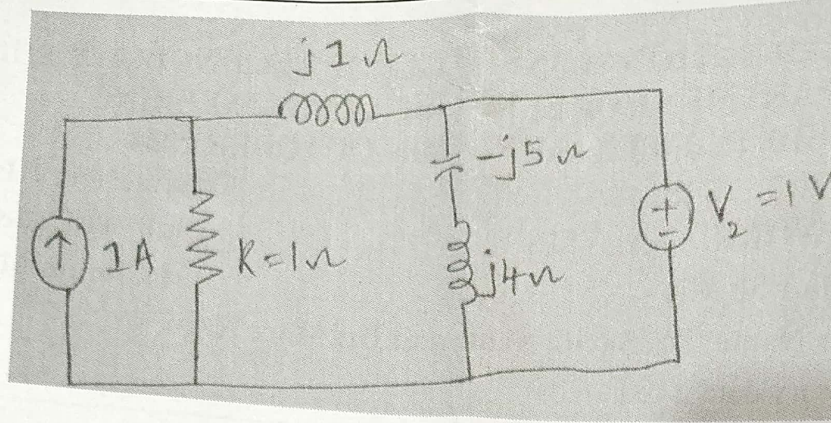
Time Duration: 1 Hour 30 Min

Maximum Marks: 20

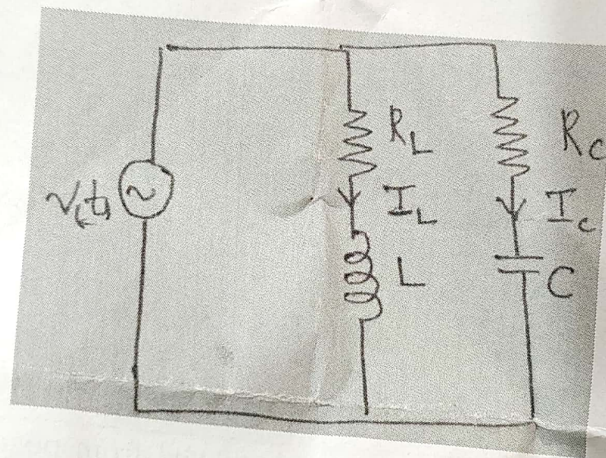
Subject Coordinator Name: Dr. Savita Nema and Dr. More Raju

Note: Assume necessary data if required.

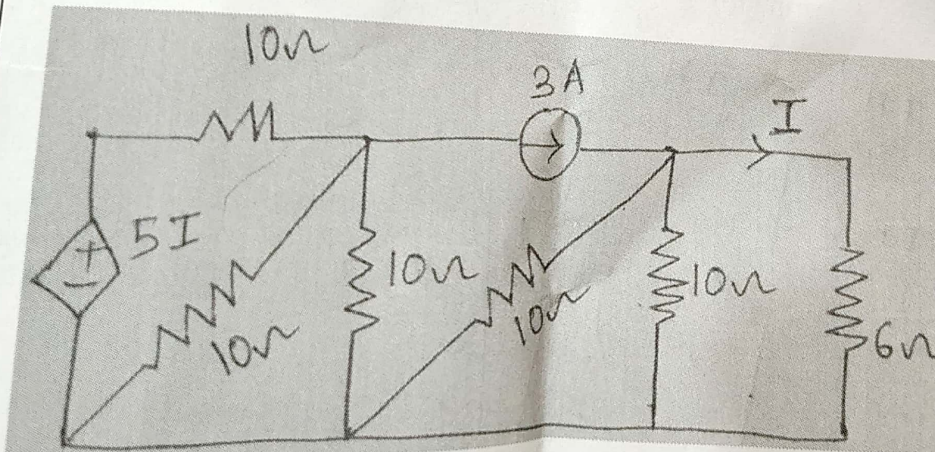
S.No	Question	Marks	CO
1	For the circuit shown below, find $i_2(t)$ using Laplace transform. Switch K is closed at $t = 0$. 	5	CO-2
2	For the circuit shown below, switch is moved from position a to position b at $t = 0$. Find $i(0^+)$, $\frac{di}{dt}(0^+)$ and $\frac{d^2i}{dt^2}(0^+)$. 	5	CO-2
3	A) Define Superposition theorem. Explain why superposition theorem is not applicable for power calculations. B) Calculate the voltage V across the resistance R in the figure shown by the principle of Superposition. Also verify the solution obtained.	2 3	CO-1



- A) For the circuit shown below, draw the phasor diagram. Derive the condition for the two branch currents I_L and I_C to be in quadrature.



- B) Determine current through the 6Ω resistor shown in figure below.



3

CO-1

2

CO-1